A final double transposition before a singleton attachment

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \ldots, \rho_{r-1} \rangle \leq \mathrm{Alt}(\Omega)$ be transitive with a string C-marking, where $r \geq 4$. Write

$$s = \rho_{r-1}, \quad t = \rho_{r-2}, \quad u = \rho_{r-3}, \qquad P = G_{\leq r-2}, \quad J = G_{\leq r-3}.$$

Assume $P = \mathrm{Alt}(X)$ is the actual natural alternating group on $X = \Omega \setminus \{z\}$. Assume also that J has an orbit C of size at least four and contains the pure natural subgroup $\mathrm{Alt}(C)$.

Theorem 1. *If t is a double transposition, the actual last three marks form the six-point A_5 marking of source Table 2(3), read from the terminal end. More precisely, there are six distinct points z, a, e, e', c, d such that*

$$s = (z\ a)(e\ e'), \qquad t = (a\ e)(e'\ c), \qquad u = (e\ e')(c\ d), \tag{1}$$

and these three marks fix every other point. No numerical rank bound or smaller-degree induction is required for this implication.

Proof. The involution s centralizes J. Transitivity of G forces $a = z^s \neq z$. Since J fixes z and commutes with s, it fixes a as well. The centralizer of the natural $\mathrm{Alt}(C)$ fixes C pointwise when $|C| \geq 4$; hence s fixes C, and $a \notin C$.

The two edges of t must connect all the J-orbits on X, because $P = \langle J, t \rangle$ is transitive there. There are therefore at most three such orbits. If all orbits other than C were singletons, s would be a nonidentity even involution on at most three points, which is impossible. Thus the orbits are exactly C, $\{a\}$ and one nontrivial orbit E. The involution s has the form

$$s = (z\ a)\delta,$$

where δ is odd, preserves E, and centralizes the transitive action of J on E. Such a centralizing nonidentity involution is free: a fixed point would, by transitivity and commutation, force every point fixed. Write $|E| = 2m$; oddness of δ says that m is odd.

Contracting the three J-orbits, the two edges of t give a tree. They cannot both meet the singleton a, since t is an involution. There are precisely two forms:

$$\begin{array}{ll} \text{I:} & t = (a\ c)(e\ c'), \quad c, c' \in C,\ c \neq c',\ e \in E; \\ \text{II:} & t = (a\ e)(f\ c), \quad c \in C,\ e, f \in E,\ e \neq f. \end{array} \tag{2}$$

Put $e' = e^\delta$. Full IP gives

$$\langle s, t \rangle \cap P = \langle t \rangle. \tag{3}$$

Every element of $\langle s, t \rangle$ fixing z is even and supported on X, so belongs to the left side of (3).

In form I, st is a three-cycle on each of $\{z, a, c\}$ and $\{c', e, e'\}$. On the remaining points it is the product of the other $m - 1$ pairs of δ. Thus $(st)^3$ is exactly that remaining product. It fixes z and a, so (3) forces it to be the identity; t moves a. Hence $m = 1$.

In form II with $f = e'$, the product st is a five-cycle on $\{z, a, e, e', c\}$ and is δ on the remaining $m - 1$ pairs. Now $(st)^5$ and (3) again give $m = 1$. If $f \neq e'$, the product st is a four-cycle on $\{z, a, e, e'\}$ and a three-cycle on $\{f, f^\delta, c\}$. Its fourth power fixes z but retains that three-cycle. It cannot belong to $\langle t \rangle$, contrary to (3). This proves, in every surviving case,

$$E = \{e, e'\}, \qquad \delta = (e\ e'). \tag{4}$$

Set $J' = G_{\leq r-4}$. It fixes a and centralizes t. Because J' preserves E and t carries a point of E outside E, that point is fixed by J': its image must lie both in E and in the inverse t-image of the other J-orbit. That intersection is a singleton in either form of (2). Hence J' fixes both points of E. Transitivity of J on E then forces u to swap e, e'.

Consider form II. Commutation with $t = (a\ e)(e'\ c)$ now also forces J' to fix c. Since J is transitive on C, u moves c to a distinct point $d \in C$. Consequently

$$u = (e\ e')(c\ d)\epsilon,$$

where ϵ is an even involution supported on $C \setminus \{c, d\}$. On the six points in (1), the three core permutations are literally the accepted Table 2(3) marking, with labels $1, \ldots, 6$ corresponding to z, a, e, e', c, d. Their group is the perfect group A_5.

Let $Q = \langle s, t, u \rangle$. Its only exterior action is $\langle \epsilon \rangle$. If $\epsilon \neq 1$ and projection to the six-point core were injective, its exterior action would define a nontrivial homomorphism $A_5 \to C_2$: the first two core marks map to 1, and the third to ϵ. Perfection excludes this. Thus $\epsilon \in Q$. As $\epsilon \in \mathrm{Alt}(C) \leq J$, full IP puts it in $Q \cap J = \langle u \rangle$. It fixes E, whereas u swaps E, so $\epsilon = 1$. We obtain exactly (1).

It remains to exclude form I. Here commutation with $t = (a\ c)(e\ c')$ forces J' to fix both c, c'. Neither can be fixed by u, nor can u exchange them: either event would create a proper J-orbit inside C. Thus u sends them to distinct $d, d' \in C \setminus \{c, c'\}$, and

$$u = (e\ e')(c\ d)(c'\ d')\epsilon,$$

where ϵ is an odd involution supported on $C \setminus \{c, c', d, d'\}$. It is disjoint from the two four-point sets

$$A = \{z, a, c, d\}, \qquad B = \{e, e', c', d'\}.$$

The core projections of Q on A and B are each S_4. On A their three marked restrictions are the successive path transpositions $(z\ a), (a\ c), (c\ d)$. On B they are $(e\ e'), (e\ c'), (e\ e')(c'\ d')$; the first and third give $(c'\ d')$, completing a path.

The word $(tu)^4$ is a pure three-cycle on A: tu has orders three and four on A and B, respectively, and its exterior factor ϵ disappears in the fourth power. Likewise $(stu)^4$ is a pure three-cycle on B: the core orders are four and three, respectively. Conjugating the first pure three-cycle by Q supplies all three-cycles on A, since its projection there is S_4; conjugating the second does the same on B. Therefore Q contains the independent pure subgroups $\mathrm{Alt}(A)$ and $\mathrm{Alt}(B)$.

The group A_4 is transitive on the six unordered pairs of its four points. Conjugate $s = (z\ a)(e\ e')$ by an element of $\mathrm{Alt}(A) \times \mathrm{Alt}(B)$ sending these pairs to $\{c, d\}$ and $\{c', d'\}$. This gives

$$v = (c\ d)(c'\ d') \in Q.$$

But $v \in \mathrm{Alt}(C) \leq J$, while $Q \cap J = \langle u \rangle$. The nonidentity v fixes E and u does not. This contradiction eliminates form I and completes the proof. □

Attribution and boundary. The literal six-point A_5 marking and its identification are borrowed from the accepted Table 2 recovery, with its complete census and source transcription retained as required inputs. The pure prefix alternating kernel is an explicit hypothesis; its acquired proof and root's initial kernel suggestion remain paid where this theorem is applied. The new argument is the orbit-tree reduction, the two-generator stabilizer test, and the two pure A_4 factors excluding the other attachment. The last-prefix double-transposition hypothesis has not been proved for every remaining odd-degree configuration. Even when it holds, the conclusion is an actual opposite head, not an exclusion of all opposite heads or a complete L044 recovery. No new computation, formal result, original or question credit is claimed.